

PERFORMANCE TESTING PHASE

Project Performance Testing

Date	01 JULY 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Introduction

Performance testing is an important phase of the project development lifecycle that evaluates whether the developed solution performs efficiently under different operating conditions. In the **Strategic Product Placement Analysis** project, testing was conducted to ensure that the Tableau dashboard accurately processes retail sales data, displays interactive visualizations correctly, and provides a smooth user experience.

The testing process verified dashboard functionality, data accuracy, loading performance, compatibility, and usability before publishing the project on Tableau Public.

Objectives of Performance Testing

- Verify the accuracy of dashboard calculations.
- Ensure dashboards load efficiently.
- Test interactive filters and visualizations.
- Validate data consistency with the original dataset.
- Confirm compatibility across different devices and browsers.
- Evaluate overall usability of the dashboard.

Testing Types

Testing Type	Purpose
Functional Testing	Verify dashboard functions correctly.
Accuracy Testing	Validate calculated values and visualizations.
Performance Testing	Measure dashboard loading and response time.
Usability Testing	Evaluate ease of use and navigation.
Compatibility Testing	Ensure dashboard works across browsers and devices.

Test Case Results

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Import CSV dataset	Dataset imported successfully	Successfully imported	 Pass
TC-02	Display Product Category chart	Chart displays correctly	Working as expected	 Pass
TC-03	Display Promotion Analysis	Promotion visualization generated	Working correctly	 Pass
TC-04	Apply Dashboard Filters	Filters update dashboard dynamically	Filters functioning properly	 Pass
TC-05	Open Tableau Story	Story pages load correctly	Successfully displayed	 Pass
TC-06	Publish on Tableau Public	Dashboard published successfully	Successfully published	 Pass
TC-07	GitHub Repository Access	Repository opens successfully	Accessible	 Pass

Accuracy Testing

Accuracy testing ensures that all dashboard values match the original dataset and that calculations are performed correctly.

Test Item	Expected Result	Status
Sales Volume	Matches dataset values	 Pass
Product Category	Correct categorization	 Pass
Promotion Analysis	Accurate calculations	 Pass
Competitor Pricing	Correct comparison	 Pass
Seasonal Analysis	Accurate visualization	 Pass
KPI Cards	Correct totals and averages	 Pass

Performance Testing

Performance testing measures how quickly the dashboard loads and responds to user interactions.

Performance Parameter	Expected Result	Observed Result	Status
Dashboard Loading	< 5 seconds	3–4 seconds	✔ Pass
Chart Rendering	Smooth visualization	Smooth	✔ Pass
Filter Response	Immediate update	Instant	✔ Pass
Story Navigation	No delay	Smooth	✔ Pass
Dashboard Refresh	Quick refresh	Successful	✔ Pass

Usability Testing

Usability testing evaluates whether users can easily understand and interact with the dashboard.

Usability Criteria	Observation	Status
Dashboard Navigation	Easy to navigate	✔ Pass
Chart Readability	Clear and understandable	✔ Pass
Filter Accessibility	User-friendly	✔ Pass
Story Presentation	Logical sequence	✔ Pass
Overall User Experience	Satisfactory	✔ Pass

Compatibility Testing

Compatibility testing verifies dashboard accessibility across multiple environments.

Platform	Result	Status
Tableau Desktop	Working correctly	✔ Pass
Tableau Public	Accessible online	✔ Pass
Google Chrome	Compatible	✔ Pass
Microsoft Edge	Compatible	✔ Pass
Windows 10/11	Fully supported	✔ Pass

Testing Summary

Testing Activity	Outcome
Functional Testing	Successfully completed
Accuracy Testing	All calculations validated
Performance Testing	Dashboard performed efficiently
Usability Testing	User-friendly interface confirmed
Compatibility Testing	Successfully tested across supported platforms

Challenges Encountered

Challenge	Resolution
Data inconsistencies	Cleaned and validated dataset before analysis
Dashboard formatting	Adjusted layouts and visualization properties
Tableau Public publishing issues	Republished after resolving connection issues
Documentation updates	Reviewed and corrected project reports

Conclusion

The **Strategic Product Placement Analysis** dashboard successfully passed all functional and performance testing activities. The dashboard accurately visualizes retail sales data, responds efficiently to user interactions, and provides a user-friendly analytical interface. Testing confirmed that the solution is reliable, accurate, compatible, and ready for deployment. The successful completion of this phase ensures that the dashboard can effectively support retail managers and business analysts in making data-driven decisions.